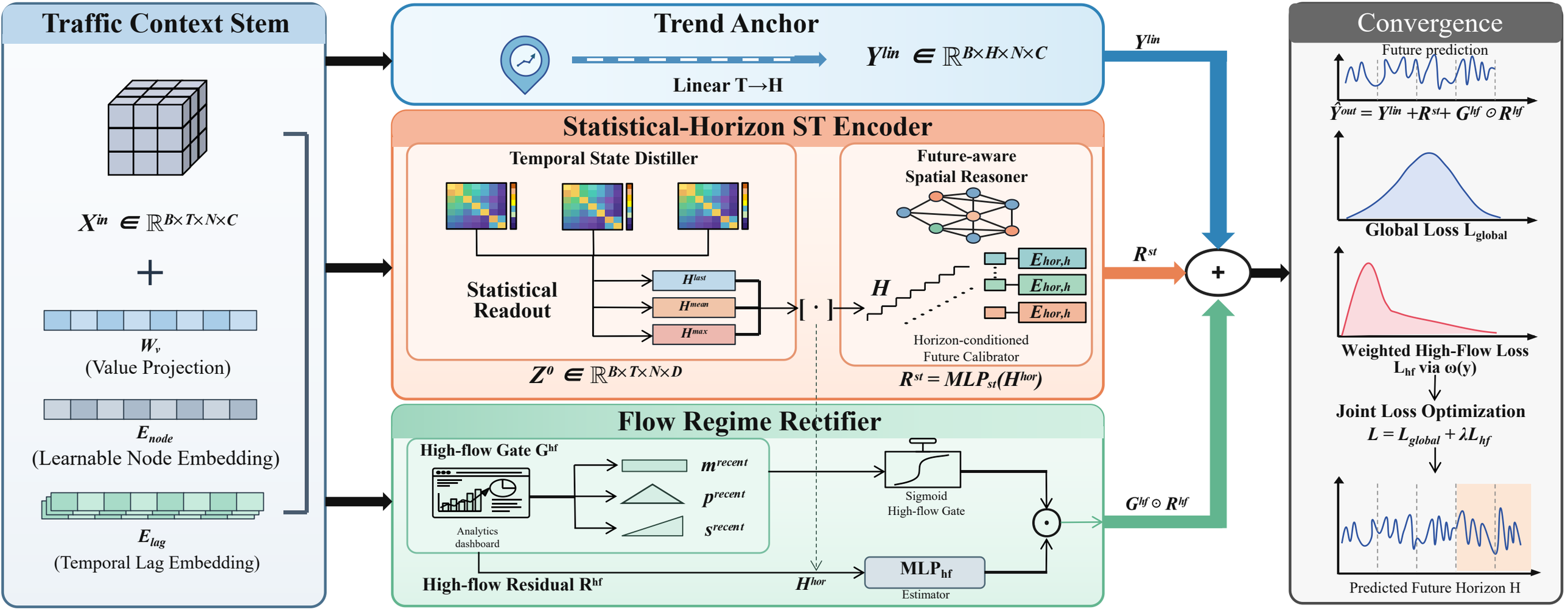

Traffic Context Stem
$X^{in} \in \mathbb{R}^{B\times T\times N\times C}$
+
W_v
(Value Projection)
E_{node}
(Learnable Node Embedding)
E_{lag}
(Temporal Lag Embedding)
Trend Anchor
Linear T→H
$Y^{lin} \in \mathbb{R}^{B\times H\times N\times C}$
Statistical-Horizon ST Encoder
Temporal State Distiller
Statistical Readout
H^{last}
H^{mean}
H^{max}
$Z^0 \in \mathbb{R}^{B\times T\times N\times D}$
Future-aware Spatial Reasoner
H
$E_{hor,h}$
$E_{hor,h}$
$E_{hor,h}$
Horizon-conditioned Future Calibrator
$R^{st} = MLP_{st}(H^{hor})$
Flow Regime Rectifier
High-flow Gate G^{hf}
Analytics dashboard
m^{recent}
p^{recent}
s^{recent}
Sigmoid High-flow Gate
High-flow Residual R^{hf}
H^{hor}
MLP_{hf}
Estimator
Y^{lin}
R^{st}
$G^{hf} \odot R^{hf}$
Convergence
Future prediction
$\hat{Y}^{out} = Y^{lin} + R^{st} + G^{hf} \odot R^{hf}$
Global Loss L_{global}
Weighted High-Flow Loss L_{hf} via ω(y)
Joint Loss Optimization
$L = L_{global} + \lambda L_{hf}$
Predicted Future Horizon H